SPREAD OPERATOR

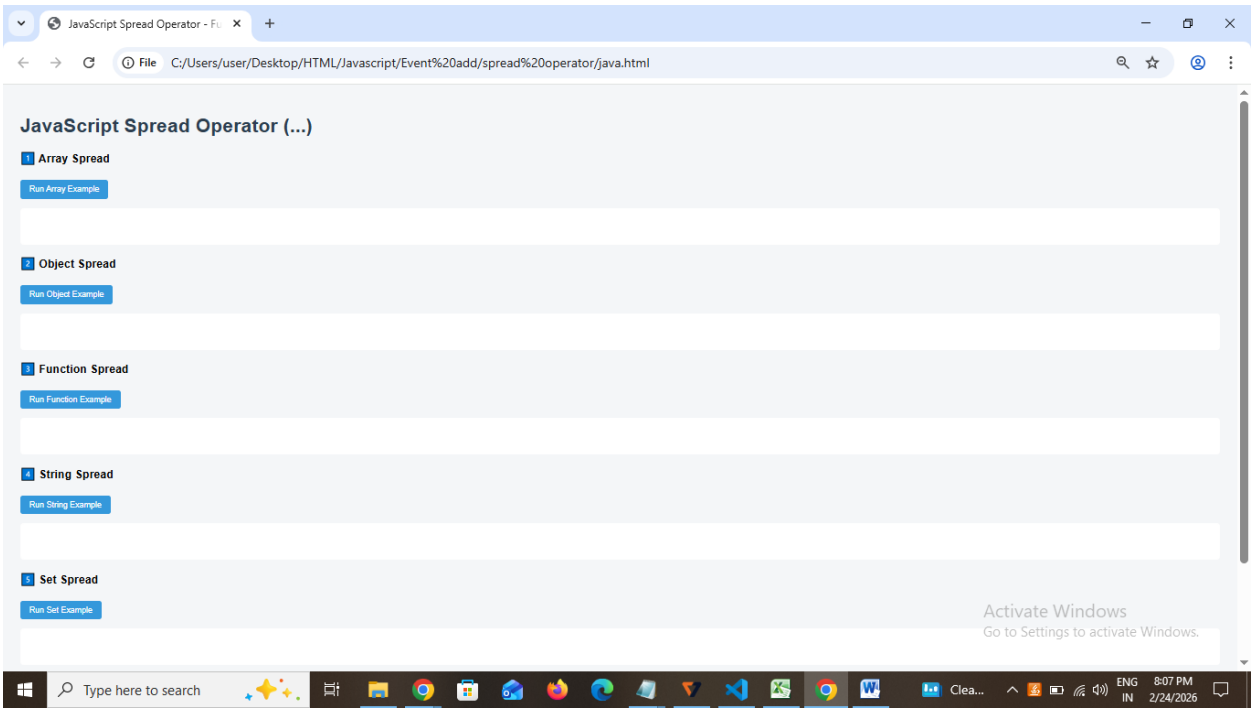
The **spread operator** in JavaScript, denoted by three dots (`...`), is used to expand elements of an iterable (like an array, string, Map, or Set) into individual elements. For objects, introduced in ES2018, it expands properties into a new object. It is widely used for copying, merging, and passing values in a concise, readable, and often immutable way.

INPUT:

```html
<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Spread Operator - Full Demo</title>
    <style>
        body {
            font-family: Arial;
            padding: 20px;
            background-color: #f4f6f8;
        }

        h1 {
            color: #2c3e50;
        }

        button {
            padding: 8px 15px;
            margin: 5px 0;
            cursor: pointer;
            border: none;
            background-color: #3498db;
            color: white;
            border-radius: 4px;
        }

        button:hover {
            background-color: #2980b9;
        }

        .output {
            background: white;
            padding: 10px;
            margin-top: 10px;
            border-radius: 5px;
            min-height: 40px;
```

```
        }
    </style>
</head>
<body>

<h1>JavaScript Spread Operator (...)</h1>

<h3>1 Array Spread</h3>
<button onclick="arraySpread()">Run Array Example</button>
<div class="output" id="arrayOutput"></div>

<h3>2 Object Spread</h3>
<button onclick="objectSpread()">Run Object Example</button>
<div class="output" id="objectOutput"></div>

<h3>3 Function Spread</h3>
<button onclick="functionSpread()">Run Function Example</button>
<div class="output" id="functionOutput"></div>

<h3>4 String Spread</h3>
<button onclick="stringSpread()">Run String Example</button>
<div class="output" id="stringOutput"></div>

<h3>5 Set Spread</h3>
<button onclick="setSpread()">Run Set Example</button>
<div class="output" id="setOutput"></div>

<h3>6 Rest vs Spread</h3>
<button onclick="restExample()">Run Rest Example</button>
<div class="output" id="restOutput"></div>

<script>

function arraySpread() {
    const arr1 = [1, 2, 3];
    const arr2 = [4, 5];

    const copy = [...arr1];
    const merged = [...arr1, ...arr2];

    document.getElementById("arrayOutput").innerHTML =
        "Original: " + arr1 +
        "<br>Copy: " + copy +
        "<br>Merged: " + merged;
}
```

```javascript
function objectSpread() {
    const user = { name: "John", age: 25 };
    const updatedUser = { ...user, age: 30, city: "New York" };

    document.getElementById("objectOutput").innerHTML =
        "Original: " + JSON.stringify(user) +
        "<br>Updated: " + JSON.stringify(updatedUser);
}

function functionSpread() {
    function sum(a, b, c) {
        return a + b + c;
    }

    const numbers = [10, 20, 30];
    const result = sum(...numbers);

    document.getElementById("functionOutput").innerHTML =
        "Numbers: " + numbers +
        "<br>Sum: " + result;
}

function stringSpread() {
    const str = "Hello";
    const letters = [...str];

    document.getElementById("stringOutput").innerHTML =
        "String: " + str +
        "<br>Letters: " + letters;
}

function setSpread() {
    const set = new Set([1, 2, 3]);
    const array = [...set];

    document.getElementById("setOutput").innerHTML =
        "Set converted to Array: " + array;
}

function restExample() {
    const arr = [1, 2, 3, 4];
    const [first, ...rest] = arr;

    document.getElementById("restOutput").innerHTML =
```

```
        "Array: " + arr +
        "<br>First: " + first +
        "<br>Rest: " + rest;
}

</script>

</body>
</html>
```

OUTPUT: